Unbiased Estimators & Confidence Intervals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Unbiased Estimator of Population Mean & Variance**

A sample of size $n = 49$ has mean $\bar{x} = 44$, drawn from a population with $\mu_0 = 47$ and known $\sigma = 10$. Find the test statistic z, correct to 2 decimal places.

Final Answer: -2.10. Working through this statistics step: $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{44 - 47}{10/\sqrt{49}} \approx -2.10$

2 **[Foundational]** • **Target: Confidence Intervals for Mean (Known Variance)**

A sample of size $n = 49$ has mean $\bar{x} = 96$, drawn from a population with $\mu_0 = 98$ and known $\sigma = 12$. Find the test statistic z, correct to 2 decimal places.

Final Answer: -1.17. Working through this statistics step: $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{96 - 98}{12/\sqrt{49}} \approx -1.17$

3 **[Foundational]** • **Target: Confidence Intervals for Mean (Unknown Variance, t-distribution)**

A sample of size $n = 25$ has mean $\bar{x} = 76$, drawn from a population with $\mu_0 = 80$ and known $\sigma = 8$. Find the test statistic z, correct to 2 decimal places.

Final Answer: -2.50. Working through this statistics step: $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{76 - 80}{8/\sqrt{25}} \approx -2.50$

4 **[Foundational]** • **Target: Interpretation of Confidence Levels**

A sample of size $n = 25$ has mean $\bar{x} = 62$, drawn from a population with $\mu_0 = 59$ and known $\sigma = 10$. Find the test statistic z, correct to 2 decimal places.

Final Answer: 1.50. Working through this statistics step: $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{62 - 59}{10/\sqrt{25}} \approx 1.50$

5 **[Foundational]** • **Target: Unbiased Estimator of Population Mean & Variance**

A sample of size $n = 25$ has mean $\bar{x} = 51$, drawn from a population with $\mu_0 = 54$ and known $\sigma = 5$. Find the test statistic z, correct to 2 decimal places.

Final Answer: -3.00. Working through this statistics step: $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{51 - 54}{5/\sqrt{25}} \approx -3.00$

6 **[Foundational]** • **Target: Confidence Intervals for Mean (Known Variance)**

A sample of size $n = 49$ has mean $\bar{x} = 87$, drawn from a population with $\mu_0 = 85$ and known $\sigma = 5$. Find the test statistic z, correct to 2 decimal places.

Final Answer: 2.80. Working through this statistics step: $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{87 - 85}{5/\sqrt{49}} \approx 2.80$

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7 [Foundational] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution)

If $X \sim N(\mu, \sigma^2)$, approximately what percentage of values lie within 3 standard deviations of the mean? Show your complete algebraic work.

Final Answer: 99.7%. Working through this statistics step:

By the empirical rule, about 99.7% of values lie within 3σ of μ.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Interpretation of Confidence Levels

If $X \sim N(\mu, \sigma^2)$, approximately what percentage of values lie within 2 standard deviations of the mean? Show your complete algebraic work.

Final Answer: 95.4%. Working through this statistics step:

By the empirical rule, about 95.4% of values lie within 2σ of μ.

9 [Intermediate] • Target: Unbiased Estimator of Population Mean & Variance

If $X \sim N(\mu, \sigma^2)$, approximately what percentage of values lie within 1 standard deviation of the mean? Show your complete algebraic work.

Final Answer: 68.3%. Working through this statistics step:

By the empirical rule, about 68.3% of values lie within 1σ of μ.

10 [Intermediate] • Target: Confidence Intervals for Mean (Known Variance)

Consider the function $f(x) = \frac{x^2 - 9}{x - k}$. The graph of $y = f(x)$ has a removable discontinuity at $x = 3$. Find the value of k. The graph of $y = f(x)$ is shown below.

Final Answer: 3. For the function $f(x) = \frac{x^2 - 9}{x - k}$ to have a removable discontinuity at $x = 3$, the numerator and the denominator must both be zero at $x = 3$. This means that $(x - 3)$ must be a factor of both the numerator and the denominator. The numerator is $x^2 - 9$, which factors as $(x - 3)(x + 3)$. For the denominator to have a factor of $(x - 3)$, we must have $x - k = 0$ when $x = 3$. Thus, $3 - k = 0$, which implies $k = 3$. When $k = 3$, $f(x) = \frac{(x - 3)(x + 3)}{x - 3}$. For $xeq 3$, $f(x) = x + 3$..

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11 [Intermediate] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution)

Consider the quadratic function $f(x) = x^2 - 8x + k$. The graph of $y = f(x)$ is a parabola that intersects the x-axis at the points $(2, 0)$ and $(6, 0)$. Find the value of k.

Final Answer: 12. Since the parabola intersects the x-axis at $(2, 0)$ and $(6, 0)$, these are the roots of the quadratic equation $f(x) = 0$. Therefore, we can write the function in factored form as $f(x) = a(x - 2)(x - 6)$. Expanding this, we get $f(x) = a(x^2 - 8x + 12)$. Comparing this with the given function $f(x) = x^2 - 8x + k$, we can see that the coefficient of x^2 is 1. Thus, $a = 1$. So, $f(x) = x^2 - 8x + 12$. By comparing the constant terms, we find that $k = 12$. Alternatively, we can substitute one of th.

12 [Intermediate] • Target: Interpretation of Confidence Levels

The diagram shows the graph of the function $f(x) = x^2 - 4x + 3$ and a line $y = g(x)$. The line $y = g(x)$ intersects the parabola $y = f(x)$ at points A and B. Point A is given as $(1, 0)$. Find the x-coordinate of point B.

Final Answer: 3. The points of intersection A and B are found by setting the two functions equal to each other: $f(x) = g(x)$. We are given $f(x) = x^2 - 4x + 3$ and from the diagram, the line is $y = x - 1$, so $g(x) = x - 1$. Setting them equal: $x^2 - 4x + 3 = x - 1$. Rearranging the equation to form a quadratic equation: $x^2 - 4x - x + 3 + 1 = 0$, which simplifies to $x^2 - 5x + 4 = 0$. We can factor this quadratic equation: $(x - 1)(x - 4) = 0$. The solutions for x are $x = 1$ and $x = 4$. We are given that p.

13 [Intermediate] • Target: Unbiased Estimator of Population Mean & Variance

The graph of the function $f(x) = x^3 - 6x^2 + 5$ is shown below. The graph has a local maximum and a local minimum. Find the x-coordinate of the local minimum.

Final Answer: 4. To find the local minimum, we first need to find the derivative of the function $f(x)$. \\ \\ $f'(x) = \frac{d}{dx}(x^3 - 6x^2 + 5) = 3x^2 - 12x$. \\ \\ Local extrema occur where $f'(x) = 0$. So, we set $3x^2 - 12x = 0$. \\ \\ Factoring out $3x$, we get $3x(x - 4) = 0$. \\ \\ This gives us two critical points: $x = 0$ and $x = 4$. \\ \\ To determine whether these points are local maxima or minima, we can use the second derivative test. \\ \\ The second derivative is.

14 [Intermediate] • Target: Confidence Intervals for Mean (Known Variance)

Consider the function $f(x) = x^2 - 5x + 6$. The graph of this function is a parabola. Determine the x-coordinates of the points where the graph intersects the x-axis.

Final Answer: $2, 3$. To find the x-coordinates where the graph intersects the x-axis, we need to find the roots of the function, i.e., where $f(x) = 0$. We set the function equal to zero: $x^2 - 5x + 6 = 0$. This is a quadratic equation that can be factored. We are looking for two numbers that multiply to 6 and add to -5. These numbers are -2 and -3. So, the equation can be factored as: $(x - 2)(x - 3) = 0$. For the product of two factors to be zero, at least one of the factors must be zero. Therefore, $x - 2 = 0$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution) • Cross-Lesson: 3.01 3D Vectors & the Cross Product**

The diagram shows the graph of the function $y = f(x)$. The equation of the function is given by $f(x) = x^2 - 4x + 3$. Identify the coordinates of the local minimum of the graph.

Final Answer: (2, -1). The function is given by $f(x) = x^2 - 4x + 3$. To find the local minimum, we first find the derivative of the function: $f'(x) = 2x - 4$. We set the derivative to zero to find the critical points: $2x - 4 = 0$, which gives $x = 2$. To confirm this is a minimum, we can find the second derivative: $f''(x) = 2$. Since $f''(2) = 2 > 0$, the critical point at $x = 2$ corresponds to a local minimum. Now we find the y-coordinate by substituting $x = 2$ into the original function.

16 **[Challenging] • Target: Interpretation of Confidence Levels • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions**

The diagram shows the graph of the function $f(x) = 2x^2 - 8x + k$. The graph intersects the x-axis at points A and B, and the vertex of the parabola is at point V. Given that the x-coordinate of the vertex is 2, and the y-coordinate of the vertex is -6, find the value of k.

Final Answer: 2. The vertex of a parabola given by $f(x) = ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$. In this case, $f(x) = 2x^2 - 8x + k$, so $a = 2$ and $b = -8$. The x-coordinate of the vertex is given as 2, which is consistent: $x = -\frac{-8}{2(2)} = \frac{8}{4} = 2$. The y-coordinate of the vertex is $f(2)$. We are given that the y-coordinate of the vertex is -6. Therefore, $f(2) = -6$. Substitute $x = 2$ into the function: $f(2) = 2(2)^2 - 8(2) + k$. $f(2) = 2(4) - 16 + k$ $f(2) = 8 - 16 + k$ $f(2) = -8 + k$. S.

17 **[Challenging] • Target: Unbiased Estimator of Population Mean & Variance • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)**

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. Determine the x-coordinate of the vertex of the parabola.

Final Answer: 2. The x-coordinate of the vertex of a parabola given by the equation $f(x) = ax^2 + bx + c$ is found using the formula $x = -\frac{b}{2a}$. In this case, $a = 1$ and $b = -4$. Therefore, the x-coordinate of the vertex is $x = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. Alternatively, the x-coordinate of the vertex is the midpoint of the roots. The roots are the x-intercepts, which can be found by setting $f(x) = 0$: $x^2 - 4x + 3 = 0$. Factoring this gives $(x - 1)(x - 3) = 0$, so the roots are $x = 1$ and $x = 3$..

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18 [Challenging] • Target: Confidence Intervals for Mean (Known Variance) • Cross-Lesson: 4.02 The Poisson Distribution

The following bar chart shows the number of students who chose each of four different subjects in a school. Let N be the total number of students surveyed. Find the value of N.

Final Answer: 100. The bar chart shows the number of students for each subject: Subject 1 has 25 students, Subject 2 has 28 students, Subject 3 has 22 students, and Subject 4 has 25 students. The total number of students surveyed, N, is the sum of the students who chose each subject. Therefore, $N = 25 + 28 + 22 + 25 = 100$. Choice A is incorrect as it is the sum of only the first three subjects and the last subject if it were 20. Choice C is incorrect as it omits some students or miscounts the values. Choice D i.

19 [Challenging] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution) • Cross-Lesson: 5.01 Further Integration Techniques

The diagram shows the graph of a quadratic function $y = f(x) = x^2 - 6x + k$. The graph intersects the x-axis at point A, and has a minimum point at B. The coordinates of point A are $(3, 0)$. Find the value of k.

Final Answer: 9. The function is given by $f(x) = x^2 - 6x + k$. We are given that the graph intersects the x-axis at point A $(3, 0)$. This means that when $x = 3$, $f(x) = 0$. Substituting these values into the function's equation: $0 = (3)^2 - 6(3) + k\ 0 = 9 - 18 + k\ 0 = -9 + k\ k = 9$ Therefore, the value of k is 9. Common algebraic traps include incorrectly substituting the coordinates or making errors in the order of operations when solving for k. For example, mistaking the point $(3, 0)$ as the vertex wo.

20 [Challenging] • Target: Interpretation of Confidence Levels • Cross-Lesson: 5.02 Differential Equations

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is a parabola. Find the coordinates of the vertex of the parabola.

Final Answer: (2, -1). The function is given by $f(x) = x^2 - 4x + 3$. This is a quadratic function in the form $ax^2 + bx + c$, where $a = 1$, $b = -4$, and $c = 3$. The x-coordinate of the vertex of a parabola is given by the formula $x = -\frac{b}{2a}$. Substituting the values of a and b, we get $x = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. To find the y-coordinate of the vertex, substitute this x-value back into the function: $f(2) = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1$. Therefore, the coordinates of the vertex are (.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Unbiased Estimator of Population Mean & Variance

Given sample observations $x_1, \ldots, x_n$, what is the formula for the unbiased estimator s^2_{n-1} of the population variance σ^2?

a. $s^2_{n-1} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$ ✓

b. $s^2_n = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$

c. $s^2 = \frac{1}{n+1}\sum_{i=1}^{n}x_i^2$

d. $s^2 = \frac{(\sum x_i)^2}{n-1}$

Dividing by $n-1$ removes the downward bias of the sample variance, ensuring that $E(s^2_{n-1}) = \sigma^2$.

22 [Foundational] • Target: Unbiased Estimator of Population Mean & Variance

A sample of $n = 5$ values has $\sum x_i = 30$ and $\sum x_i^2 = 210$. Calculate the unbiased estimate of the population variance s^2_{n-1}.

a. 7.5 ✓

b. 6.0

c. 30

d. 9.0

$\bar{x} = 30/5 = 6$. $\sum(x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2 = 210 - 5(36) = 210 - 180 = 30$. Unbiased variance: $s^2_{n-1} = \frac{30}{5-1} = \frac{30}{4} = 7.5$.

23 [Foundational] • Target: Unbiased Estimator of Population Mean & Variance

If a sample standard deviation computed with denominator n is $s_n = 4.0$ for $n = 16$, what is the unbiased standard deviation s_{n-1}?

a. $s_{n-1} = 4.0 \times \sqrt{\frac{16}{15}} \approx 4.13$ ✓

b. $s_{n-1} = 4.0 \times \sqrt{\frac{15}{16}} \approx 3.87$

c. $s_{n-1} = 4.0$

d. $s_{n-1} = 4.0 \times \frac{16}{15} \approx 4.27$

$s^2_{n-1} = \frac{n}{n-1}s^2_n = \frac{16}{15}(16) \Rightarrow s_{n-1} = 4\sqrt{\frac{16}{15}} \approx 4.13$.

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24 [Foundational] • Target: Confidence Intervals for Mean (Known Variance)

A random sample of $n = 64$ items from a population with known $\sigma = 12$ has mean $\bar{x} = 50$. What is the 95% confidence interval for μ ($z^* = 1.96$)?

a. $[47.06, 52.94]$ ✓

b. $[48.50, 51.50]$

c. $[44.12, 55.88]$

d. $[49.63, 50.37]$

Standard error is $\frac{\sigma}{\sqrt{n}} = \frac{12}{\sqrt{64}} = \frac{12}{8} = 1.5$. Margin of error is $1.96(1.5) = 2.94$. Interval: $50 \pm 2.94 = [47.06, 52.94]$.

25 [Foundational] • Target: Confidence Intervals for Mean (Known Variance)

To reduce the margin of error of a confidence interval by a factor of 3, the sample size must be multiplied by:

a. 9 ✓

b. 3

c. $\sqrt{3}$

d. 6

Since margin of error is proportional to $\frac{1}{\sqrt{n}}$, to reduce it by 3 requires $\sqrt{n_{\text{new}}} = 3\sqrt{n} \Rightarrow n_{\text{new}} = 9n$.

26 [Foundational] • Target: Confidence Intervals for Mean (Known Variance)

What is the critical value z^* for a 99% confidence interval?

a. 2.576 ✓

b. 1.960

c. 1.645

d. 2.326

For a 99% two-tailed interval, each tail contains 0.005 of the probability. The corresponding standard normal critical value is $z^* \approx 2.576$.

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27 [Foundational] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution)

When constructing a confidence interval for μ with unknown population variance σ^2 from a normal sample of size $n = 10$, which distribution is used?

a. Student's t-distribution with 9 degrees of freedom ✓

b. Standard Normal Z-distribution

c. Student's t-distribution with 10 degrees of freedom

d. Chi-square distribution with 9 degrees of freedom

Replacing σ with sample standard deviation s_{n-1} necessitates the t-distribution with degrees of freedom $df = n - 1 = 10 - 1 = 9$.

28 [Foundational] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution)

How does the critical value t^* compare to z^* for the same confidence level and finite sample size?

a. $t^* > z^*$, because the t-distribution has heavier tails to reflect estimation uncertainty ✓

b. $t^* < z^*$, because t is more compact

c. $t^* = z^*$ always

d. t^* can be smaller or larger depending on $\bar{x}$

The t-distribution exhibits greater tail dispersion than the standard normal distribution to account for estimating σ from sample data, making $t^* > z^*$ for any finite degrees of freedom.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution)

A sample of $n = 16$ has $\bar{x} = 22.0$ and $s_{n-1} = 2.0$. Given $t_{0.025,\,15} = 2.131$, find the 95% confidence interval for μ.

a. $[20.93, 23.07]$ ✓

b. $[21.47, 22.53]$

c. $[19.87, 24.13]$

d. $[20.00, 24.00]$

Standard error $= \frac{s}{\sqrt{n}} = \frac{2.0}{4} = 0.50$. Margin of error $= 2.131(0.50) = 1.0655$. Interval: $22.0 \pm 1.07 = [20.93, 23.07]$.

30 [Intermediate] • Target: Interpretation of Confidence Levels

Which statement correctly conveys the meaning of a '95% confidence level'?

a. **If many random samples are drawn and intervals constructed, 95% of them will contain the true population parameter** ✓

b. There is a 95% probability that the true parameter lies inside the specific interval $[47, 53]$

c. 95% of sample data values lie between 47 and 53

d. The sample mean has a 95% chance of being correct

The confidence level describes the long-run coverage rate of the interval estimation process over repeated random sampling.

31 [Intermediate] • Target: Interpretation of Confidence Levels

If a 95% confidence interval for μ is $[102, 118]$, what is the conclusion for the two-tailed test $H_0 : \mu = 100$ vs $H_1 : \mu \neq 100$ at $\alpha = 0.05$?

a. **Reject H_0 at the 5% level, because 100 lies outside the 95% confidence interval** ✓

b. Fail to reject H_0, because 100 is close to 102

c. Accept H_0 as proven

d. The test is inconclusive without a P-value

A hypothesized value outside a $(1 - \alpha)$ confidence interval is rejected at significance level α in a two-tailed test.

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32 [Intermediate] • Target: Interpretation of Confidence Levels

What happens to the width of a confidence interval if the confidence level is increased from 90% to 99% (with sample size fixed)?

a. The interval widens ✓

b. The interval narrows

c. The interval width remains identical

d. The interval shifts to the right

Higher confidence requires capturing a larger portion of the sampling distribution, which increases the critical value and widens the margin of error.

33 [Intermediate] • Target: Unbiased Estimator of Population Mean & Variance

Consider the function $f(x) = x^2 - 6x + 8$. The graph of $y = f(x)$ intersects the x-axis at points A and B, and the y-axis at point C. Find the length of the line segment AB.

a. 2 units ✓

b. 3 units

c. 4 units

d. 6 units

To find the points where the graph intersects the x-axis, we set $f(x) = 0$. $x^2 - 6x + 8 = 0$. Factoring the quadratic equation, we get $(x - 2)(x - 4) = 0$. Thus, the x-intercepts are $x = 2$ and $x = 4$. Let these points be A and B. The distance between A and B is $|4 - 2| = 2$ units. To find the y-intercept, we set $x = 0$. $f(0) = 0^2 - 6(0) + 8 = 8$. So, point C is (0, 8). The question asks for the length of the line segment AB, which is the distance between the x-intercepts.

34 [Intermediate] • Target: Confidence Intervals for Mean (Known Variance)

The diagram shows the graph of the function $f(x) = x^2 - 6x + k$. The vertex of the parabola is at the point $(h, -1)$. Determine the value of k.

a. 8 ✓

b. 7

c. 6

d. 5

The given function is $f(x) = x^2 - 6x + k$. The x-coordinate of the vertex of a parabola in the form $ax^2 + bx + c$ is given by $h = -\frac{b}{2a}$. In this case, $a = 1$ and $b = -6$, so $h = -\frac{-6}{2(1)} = 3$. The vertex is given as $(h, -1)$, so we know $h = 3$. To find k, we substitute the coordinates of the vertex into the function: $f(3) = -1$. $(3)^2 - 6(3) + k = -1$ $9 - 18 + k = -1$ $-9 + k = -1$ $k = -1 + 9$ $k = 8$. Therefore, the value of k is 8.

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35 [Intermediate] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution)

The graph of the function $g(x) = x^2 - 6x + k$ is shown below. The graph intersects the x-axis at points P and Q. Point P has coordinates $(2, 0)$. Find the value of the parameter k.

a. 1

b. **2 ✓**

c. 4

d. 8

The graph of the quadratic function $g(x) = x^2 - 6x + k$ intersects the x-axis at points where $g(x) = 0$. We are given that one of these intersection points is P, with coordinates $(2, 0)$. This means that when $x = 2$, $g(x) = 0$. We can substitute these values into the function to find the value of k: $g(2) = (2)^2 - 6(2) + k$ $0 = 4 - 12 + k$ $0 = -8 + k$ $k = 8$ Therefore, the value of the parameter k is 8.

36 [Intermediate] • Target: Interpretation of Confidence Levels

Consider the graph of the quadratic function $y = x^2 - 4x + 3$. What are the coordinates of the vertex of this parabola?

a. **(2, -1) ✓**

b. (1, 0)

c. (3, 0)

d. (2, 1)

The given quadratic function is in the form $y = ax^2 + bx + c$, where $a = 1, b = -4, c = 3$. The x-coordinate of the vertex of a parabola is given by the formula $x = -b/(2a)$. In this case, $x = -(-4)/(2 \times 1) = 4/2 = 2$. To find the y-coordinate of the vertex, substitute this x-value back into the function: $y = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1$. Therefore, the coordinates of the vertex are (2, -1).

37 [Intermediate] • Target: Unbiased Estimator of Population Mean & Variance

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B. Determine the distance between A and B.

a. **2 ✓**

b. 3

c. 4

d. 5

To find the x-intercepts, we set $f(x) = 0$. Thus, we solve the quadratic equation $x^2 - 4x + 3 = 0$. This can be factored as $(x - 1)(x - 3) = 0$. The solutions are $x = 1$ and $x = 3$. These are the x-coordinates of points A and B. Therefore, the points are (1, 0) and (3, 0). The distance between these two points on the x-axis is the absolute difference of their x-coordinates: $|3 - 1| = 2$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Confidence Intervals for Mean (Known Variance) • Cross-Lesson: 4.02 The Poisson Distribution

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. Determine the value of $f'(2)$.

a. -2

b. **0** ✓

c. 2

d. 4

The function is given by $f(x) = -x^2 + 4x - 3$. To find the derivative $f'(x)$, we differentiate each term with respect to x: $f'(x) = \frac{d}{dx}(-x^2) + \frac{d}{dx}(4x) - \frac{d}{dx}(3)$ $f'(x) = -2x + 4 - 0$ $f'(x) = -2x + 4$. To find the value of $f'(2)$, we substitute $x = 2$ into the derivative: $f'(2) = -2(2) + 4$ $f'(2) = -4 + 4$ $f'(2) = 0$. Therefore, the value of $f'(2)$ is 0.

39 [Challenging] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution) • Cross-Lesson: 5.01 Further Integration Techniques

Consider the quadratic function $f(x) = x^2 - 4x + 3$. Find the value of x where the graph of $f(x)$ intersects the x-axis at the point $(x, 0)$, given that $x > 1$.

a. 2

b. **3** ✓

c. 4

d. 5

The graph of $f(x)$ intersects the x-axis when $f(x) = 0$. We set the function equal to zero and solve for x: $x^2 - 4x + 3 = 0$ Factoring the quadratic equation, we get: $(x - 1)(x - 3) = 0$ This gives two possible solutions for x: $x = 1$ or $x = 3$. The question specifies that we need to find the value of x where $x > 1$. Therefore, the correct value is $x = 3$.

40 [Challenging] • Target: Interpretation of Confidence Levels • Cross-Lesson: 5.02 Differential Equations

The diagram shows the graph of the function $f(x) = 3x^2 + bx - 6$. The graph passes through the point $(1, 1)$. Find the value of b.

a. 3

b. **4** ✓

c. 5

d. 6

The function is given by $f(x) = 3x^2 + bx - 6$. Since the graph passes through the point $(1, 1)$, we can substitute $x = 1$ and $f(x) = 1$ into the equation: $1 = 3(1)^2 + b(1) - 6$ $1 = 3 + b - 6$ $1 = b - 3$ Adding 3 to both sides, we get: $b = 1 + 3$ $b = 4$

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41 [Challenging] • Target: Unbiased Estimator of Population Mean & Variance • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B, and the y-axis at point C. Find the coordinates of the point where the graph of $y = f(x)$ intersects the y-axis.

a. **(0, 3)** ✓

b. (3, 0)

c. (1, 0)

d. (0, -3)

To find the y-intercept, we need to find the value of $f(x)$ when $x = 0$. \\ Substituting $x = 0$ into the function: \\ \(f(0) = (0)^2 - 4(0) + 3 \\ f(0) = 0 - 0 + 3 \\ f(0) = 3 \\ Therefore, the graph intersects the y-axis at the point (0, 3).

42 [Challenging] • Target: Confidence Intervals for Mean (Known Variance) • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A particle's position in the complex plane is given by $z(t) = (2t^2 - 3t + 1) + i(t^3 - 6t^2 + 11t - 6)$, where t is time in seconds and $t \geq 0$. The particle starts at $t = 0$ and its path is traced by the complex number $z(t)$. The particle is said to be 'at rest' when its velocity, represented by the complex number $v(t) = z'(t)$, is zero. Determine the times $t > 0$ when the particle is at rest.

a. $t = 1$ and $t = 3$

b. $t = 1$ and $t = 2$

c. $t = 2$ and $t = 3$

d. $t = 1, t = 2,$ **and** $t = 3$ ✓

The particle's position is given by $z(t) = (2t^2 - 3t + 1) + i(t^3 - 6t^2 + 11t - 6)$. The velocity is the derivative of the position with respect to time, $v(t) = z'(t)$. Differentiating the real and imaginary parts separately: Real part of $z'(t)$: $\frac{d}{dt}(2t^2 - 3t + 1) = 4t - 3$ Imaginary part of $z'(t)$: $\frac{d}{dt}(t^3 - 6t^2 + 11t - 6) = 3t^2 - 12t + 11$ So, $v(t) = (4t - 3) + i(3t^2 - 12t + 11)$. The particle is at rest when $v(t) = 0$. This means both the real and imaginary parts of $v(t)$ must be.

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43 [Challenging] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution) • Cross-Lesson: 6.01 Matrices (AI HL)

Consider the function $f(x) = \frac{ax^2 + bx + c}{x - d}$, where a, b, c, and d are constants. The graph of $f(x)$ has a vertical asymptote at $x = 3$ and an oblique asymptote $y = 2x + 1$. Given that $f(1) = -4$ and $f(2) = -6$, find the value of a.

a. -2

b. 2 ✓

c. -4

d. 4

The vertical asymptote occurs where the denominator is zero, so $x - d = 0$. Given the vertical asymptote is at $x = 3$, we have $d = 3$. Thus, $f(x) = \frac{ax^2 + bx + c}{x - 3}$. For the oblique asymptote, we perform polynomial long division or consider the limit as $x \to \infty$. The oblique asymptote is $y = 2x + 1$. This implies that as $x \to \infty$, $\frac{ax^2 + bx + c}{x - 3} \approx 2x + 1$. This can be seen by dividing the numerator by the denominator: $ax + b + \frac{c'}{x - 3}$. For the oblique asymptot.

44 [Challenging] • Target: Interpretation of Confidence Levels • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Consider the complex number w such that $w = x + yi$, where x and y are real numbers. Given that $w^3 = -64i$ and the imaginary part of w is positive, find the value of $x^2 + y^2$.

a. **16** ✓

b. 32

c. 64

d. 8

We are given that $w^3 = -64i$. We can express $-64i$ in polar form. The modulus is $|-64i| = 64$. The argument is $\arg(-64i) = \frac{3\pi}{2} + 2k\pi$ for integer k. Thus, $-64i = 64e^{i(\frac{3\pi}{2} + 2k\pi)}$. Let $w = re^{i\theta}$. Then $w^3 = r^3e^{i3\theta}$. Equating the moduli, we have $r^3 = 64$, so $r = 4$ (since r must be real and non-negative). Equating the arguments, we have $3\theta = \frac{3\pi}{2} + 2k\pi$. Therefore, $\theta = \frac{\pi}{2} + \frac{2k\pi}{3}$. For $k = 0$.

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45 [Challenging] • Target: Unbiased Estimator of Population Mean & Variance • Cross-Lesson: 6.03 Markov Chains (AI HL)

Consider the function $f(x) = ax^2 + bx + c$. The graph of $f(x)$ is a parabola that passes through the points $(-1, 12)$, $(1, 2)$, and $(2, 9)$. Find the value of the parameter b.

a. -3 ✓

b. 3

c. -5

d. 5

Let the function be $f(x) = ax^2 + bx + c$. We are given three points that lie on the graph of $f(x)$: $(-1, 12)$, $(1, 2)$, and $(2, 9)$. Substituting these points into the function, we get a system of three linear equations with three unknowns a, b, and c: 1. For $(-1, 12)$: $a(-1)^2 + b(-1) + c = 12 \Rightarrow a - b + c = 12$ 2. For $(1, 2)$: $a(1)^2 + b(1) + c = 2 \Rightarrow a + b + c = 2$ 3. For $(2, 9)$: $a(2)^2 + b(2) + c = 9 \Rightarrow 4a + 2b + c = 9$ We can solve this system of equations. Subtract.

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